



Experiment NO. 2

- Q. Implement & Demonstrate Declaring, Defining & Involving a simple PL/SQL function for the statement.
- i) PL/SQL function to find factorial of a number

- Declaring: function name, parameters & return type
- Defining: Write the logic inside the function body
- Involving: Call the function from a PL/SQL block & display result.

Step 1: Declare & Define the function

```
CREATE OR REPLACE FUNCTION factorial (n NUMBER)
RETURN NUMBER
IS
```

```
    fact NUMBER := 1;
```

```
BEGIN
```

```
    For i in 1...n loop
```

```
        fact := fact * i;
```

```
    END LOOP;
```

```
    RETURN fact;
```

```
END;
```

```
/
```

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Step 2: Invoke (call) the function

DECLARE

num NUMBER := 5

result NUMBER;

BEGIN

result := Factorial (num);

DBMS_OUTPUT.PUT_LINE('Factorial of ' || num ||
is : ' || result);

END;

/

OUTPUT

Factorial of 5 is : 120

SQL Plus

X

SQL*Plus: Release 21.0.0.0.0 - Production on Fri Jul 31 11:34:03 2026
Version 21.0.0.0.0

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Enter user-name: system

Enter password:

Last Successful login time: Thu Jul 30 2025 10:13:05 +05:30

Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production

Version 21.0.0.0.0

SQL> -- Create Function
SQL> CREATE OR REPLACE FUNCTION factorial(n NUMBER)

2 RETURN NUMBER

3 IS

4 fact NUMBER := 1;

5 BEGIN

6 IF n = 0 OR n = 1 THEN

7 RETURN 1;

8 END IF;

9

10 FOR i IN 2..n LOOP

11 fact := fact * i;

12 END LOOP;

13

14 RETURN fact;

15 END;

16 /

17

Function created.

SQL> SET SERVEROUTPUT ON;

SQL>

SQL> DECLARE

2 num NUMBER := 5;

3 result NUMBER;

4 BEGIN

5 result := factorial(num);

6 DBMS_OUTPUT.PUT_LINE('factorial of ' || num || ' is ' || result);

7 END;

8 /

factorial of 5 is 120

PL/SQL procedure successfully completed.

SQL>

SQL>

SQL>